BANK LOAN ANALYSIS REPORT

1. Project Overview

This project focuses on analyzing bank loan data to evaluate lending performance, borrower behavior, and credit risk patterns. The primary objective is to generate actionable insights that can help financial institutions improve decision-making in loan approvals, risk management, and portfolio optimization.

The analysis aims to:

- Identify trends in loan applications and funding
- Evaluate repayment behavior and loan performance
- Understand borrower demographics and financial profiles
- Measure risk through loan status and debt indicators

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2. Dataset Summary

- **Total Records:** 38,576
- **Total Features:** 24

Key Data Categories:

Loan Information: - Loan Amount - Interest Rate - Installment - Loan Term - Loan Purpose - Grade & Sub-grade

Borrower Details: - Annual Income - Employment Length - Home Ownership - State

Loan Performance: - Loan Status (Fully Paid, Current, Charged Off)

Payment Data: - Total Payment Received - Last Payment Date - Next Payment Date

Verification: - Verification Status

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3. Data Cleaning & Preparation (Python)

Data preprocessing was performed using Python to ensure accuracy and consistency before analysis.

Key Steps:

- **Data Loading:** Imported dataset using Pandas.
- **Initial Exploration:** Used `info()` and `describe()` to understand structure and distributions.

- **Date Standardization:** Converted all date fields into consistent datetime format.
- **Data Cleaning:**
  - Removed extra spaces from text columns
  - Standardized categorical values
- **Feature Engineering:**
  - Converted employment length into numeric format
- **Missing Values Handling:**
  - Filled missing `emp_title` values with "Unknown"
- **Duplicate Check:** Ensured dataset contained no duplicate records
- **Data Consistency:** Standardized `purpose` column naming format
- **Database Integration:** Cleaned dataset exported to MySQL for further analysis

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4. Data Analysis (SQL)

A. BANK LOAN REPORT | SUMMARY

Total Loan Applications

```
SELECT COUNT(id) AS Total_Applications

FROM bank_loan_data;
```

Total_Applications
38576

MTD Loan Applications

```
SELECT COUNT(id) AS MTD_Total_Applications

FROM bank_loan_data

WHERE MONTH(issue_date) = 12;
```

MTD_Total_Applications
4314

PMTD Loan Applications

```
SELECT COUNT(id) AS PMTD_Total_Applications
FROM bank_loan_data
WHERE MONTH(issue_date) = 11;
```

PMTD_Total_Applications
4035

Total Funded Amount

```
SELECT SUM(loan_amount) AS Total_Funded_Amount
FROM bank_loan_data;
```

Total_Funded_Amount
435757075

MTD Total Funded Amount

```
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 12;
```

MTD_Total_Funded_Amount
53981425

PMTD Total Funded Amount

```
SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 11;
```

PMTD_Total_Funded_Amount
47754825

Total Amount Received

```
SELECT SUM(total_payment) AS Total_Amount_Received

FROM bank_loan_data;
```

Total_Amount_Received
473070933

MTD Total Amount Received

```
SELECT SUM(total_payment) AS MTD_Total_Amount_Received

FROM bank_loan_data

WHERE MONTH(issue_date) = 12;
```

MTD_Total_Amount_Received
58074380

PMTD Total Amount Received

```
 SELECT SUM(total_payment) AS PMTD_Total_Amount_Received

FROM bank_loan_data

WHERE MONTH(issue_date) = 11;
```

PMTD_Total_Amount_Received
50132030

Average Interest Rate

```
SELECT AVG(int_rate) * 100 AS Avg_Int_Rate

FROM bank_loan_data;
```

Avg_Int_Rate
12.048831397760178

MTD Average Interest

```
SELECT AVG(int_rate) * 100 AS MTD_Avg_Int_Rate
FROM bank_loan_data
WHERE MONTH(issue_date) = 12;
```

	MTD_Avg_Int_Rate
▶	12.356040797403738

PMTD Average Interest

```
SELECT AVG(int_rate) * 100 AS PMTD_Avg_Int_Rate
FROM bank_loan_data
WHERE MONTH(issue_date) = 11;
```

	PMTD_Avg_Int_Rate
▶	11.941717472118796

Avg DTI

```
SELECT AVG(dti) * 100 AS Avg_DTI
FROM bank_loan_data;
```

	Avg_DTI
▶	13.32743311903776

MTD Avg DTI

```
SELECT AVG(dti) * 100 AS MTD_Avg_DTI
FROM bank_loan_data
WHERE MONTH(issue_date) = 12;
```

	MTD_Avg_DTI
▶	13.66553778395922

PMTD Avg DTI

```
SELECT AVG(dti) * 100 AS PMTD_Avg_DTI
FROM bank_loan_data
WHERE MONTH(issue_date) = 11;
```

	PMTD_Avg_DTI
▶	13.302733581164853

GOOD LOAN ISSUED

Good Loan Percentage

```
SELECT
    (COUNT(CASE WHEN loan_status IN ('Fully Paid', 'Current') THEN id END) *
100.0) /
    COUNT(id) AS Good_Loan_Percentage
FROM bank_loan_data;
```

Good_Loan_Percentage
86.17534

Good Loan Applications

```
SELECT COUNT(id) AS Good_Loan_Applications
FROM bank_loan_data
WHERE loan_status IN ('Fully Paid', 'Current');
```

Good_Loan_Applications
33243

Good Loan Funded Amount

```
SELECT SUM(loan_amount) AS Good_Loan_Funded_Amount
FROM bank_loan_data
WHERE loan_status IN ('Fully Paid', 'Current');
```

Good_Loan_Funded_Amount
370224850

Good Loan Amount Received

```
SELECT SUM(total_payment) AS Good_Loan_Amount_Received
FROM bank_loan_data
WHERE loan_status IN ('Fully Paid', 'Current');
```

Good_Loan_Amount_Received
435786170

BAD LOAN ISSUED

Bad Loan Percentage

```
SELECT
    (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0) /
    COUNT(id) AS Bad_Loan_Percentage
FROM bank_loan_data;
```

Bad_Loan_Percentage
13.824657818332

Bad Loan Applications

```
SELECT COUNT(id) AS Bad_Loan_Applications
FROM bank_loan_data
WHERE loan_status = 'Charged Off';
```

Bad_Loan_Applications
5333

Bad Loan Funded Amount

```
SELECT SUM(loan_amount) AS Bad_Loan_Funded_Amount
FROM bank_loan_data
WHERE loan_status = 'Charged Off';
```

Bad_Loan_Funded_amount
65532225

Bad Loan Amount Received

```
SELECT SUM(total_payment) AS Bad_Loan_Amount_Received
FROM bank_loan_data
WHERE loan_status = 'Charged Off';
```

Bad_Loan_amount_received
37284763

LOAN STATUS

```
SELECT
    loan_status,
    COUNT(id)           AS Loan_Count,
    SUM(total_payment)  AS Total_Amount_Received,
    SUM(loan_amount)    AS Total_Funded_Amount,
    AVG(int_rate * 100) AS Avg_Interest_Rate,
```

```
    AVG(dti * 100)       AS Avg_DTI
FROM bank_loan_data
GROUP BY loan_status;
```

loan_status	Loan_Count	Total_Amount_Received	Total_Funded_Amount	Avg_Interest_Rate	Avg_DTI
Charged Off	5333	37284763	65532225	13.878574910931917	14.004732795799695
Fully Paid	32145	411586256	351358350	11.641070773058658	13.167350754394164
Current	1098	24199914	18866500	15.0993260473588	14.724344262295068

MTD breakdown

```
SELECT
    loan_status,
    SUM(total_payment) AS MTD_Total_Amount_Received,
    SUM(loan_amount)   AS MTD_Total_Funded_Amount
FROM bank_loan_data
WHERE MONTH(issue_date) = 12
GROUP BY loan_status;
```

loan_status	MTD_Total_Amount_Received	MTD_Total_Funded_Amount
Fully Paid	47815851	41302025
Charged Off	5324211	8732775
Current	4934318	3946625

B. BANK LOAN REPORT | OVERVIEW

MONTH

```
SELECT
    MONTH(issue_date)        AS Month_Number,
    MONTHNAME(issue_date)    AS Month_Name,
    COUNT(id)                AS Total_Loan_Applications,
    SUM(loan_amount)         AS Total_Funded_Amount,
    SUM(total_payment)       AS Total_Amount_Received
FROM bank_loan_data
GROUP BY MONTH(issue_date), MONTHNAME(issue_date)ORDER BY MONTH(issue_date);
```

Month_Number	Month_Name	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	January	2332	25031650	27578836
2	February	2279	24647825	27717745
3	March	2627	28875700	32264400
4	April	2755	29800800	32495533
5	May	2911	31738350	33750523
6	June	3184	34161475	36164533
7	July	3366	35813900	38827220
8	August	3441	38149600	42682218
9	September	3536	40907725	43983948
10	October	3796	44893800	49399567
11	November	4035	47754825	50132030
12	December	4314	53981425	58074380

STATE

```
SELECT
    address_state          AS State,
    COUNT(id)              AS Total_Loan_Applications,
    SUM(loan_amount)       AS Total_Funded_Amount,
    SUM(total_payment)     AS Total_Amount_Received
FROM bank_loan_data
GROUP BY address_state
ORDER BY address_state;
```

	State	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	AK	78	1031800	1108570
2	AL	432	4949225	5492272
3	AR	236	2529700	2777875
4	AZ	833	9206000	10041986
5	CA	6894	78484125	83901234
6	CO	770	8976000	9845810
7	CT	730	8435575	9357612
8	DC	214	2652350	2921854
9	DE	110	1138100	1269136
10	FL	2773	30046125	31601905
11	GA	1355	15480325	16728040
12	HI	170	1850525	2080184
13	IA	5	56450	64482
14	ID	6	59750	65329
15	IL	1486	17124225	18875941
16	IN	9	86225	85521
17	KS	260	2872325	3247394
18	KY	320	3504100	3792530
19	LA	426	4498900	5001160
20	MA	1310	15051000	16676279
21	MD	1027	11911400	12985170
22	ME	3	9200	10808
23	MI	685	7829900	8543660
24	MN	592	6302600	6750746
25	MO	660	7151175	7692732
26	MS	19	139125	149342
27	MT	79	829525	892047
28	NC	759	8787575	9534813
29	NE	5	31700	24542
30	NH	161	1917900	2101386
31	NJ	1822	21657475	23425159
32	NM	183	1916775	2084485
33	NV	482	5307375	5451443
34	NY	3701	42077050	46108181
35	OH	1188	12991375	14330148
36	OK	293	3365725	3712649
37	OR	436	4720150	4966903
38	PA	1482	15826525	17462908
39	RI	196	1883025	2001774

TERM

```
SELECT
    term                    AS Term,
    COUNT(id)               AS Total_Loan_Applications,
    SUM(loan_amount)        AS Total_Funded_Amount,
    SUM(total_payment)      AS Total_Amount_Received
FROM bank_loan_data
GROUP BY term
ORDER BY term;
```

	Term	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
1	36 months	28237	273041225	294709458
2	60 months	10339	162715850	178361475

EMPLOYEE LENGTH

```
SELECT
    emp_length              AS Employee_Length,
    COUNT(id)               AS Total_Loan_Applications,
    SUM(loan_amount)        AS Total_Funded_Amount,
    SUM(total_payment)      AS Total_Amount_Received
FROM bank_loan_data
GROUP BY emp_length
ORDER BY emp_length;
```

Employee_Length	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
0	4575	44210625	47545011
1	3229	32883125	35498348
2	4382	44967975	49206961
3	4088	43937850	47551832
4	3428	37600375	40964850
5	3273	36973625	40397571
6	2228	25612650	27908658
7	1772	20811725	22584136
8	1476	17558950	19025777
9	1255	15084225	16516173
10	8870	116115950	125871616

PURPOSE

```
SELECT
    purpose                 AS Purpose,
    COUNT(id)               AS Total_Loan_Applications,
    SUM(loan_amount)        AS Total_Funded_Amount,
    SUM(total_payment)      AS Total_Amount_Received
FROM bank_loan_data
GROUP BY purpose
ORDER BY purpose;
```

PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
car	1497	10223575	11324914
credit card	4998	58885175	65214084
Debt consolidation	18214	232459675	253801871
educational	315	2161650	2248380
home improvement	2876	33350775	36380930
house	366	4824925	5185538
major purchase	2110	17251600	18676927
medical	667	5533225	5851372
moving	559	3748125	3999899
other	3824	31155750	33289676
renewable_energy	94	845750	898931
small business	1776	24123100	23814817
vacation	352	1967950	2116738
wedding	928	9225800	10266856

HOME OWNERSHIP

```
SELECT
    home_ownership          AS Home_Ownership,
    COUNT(id)               AS Total_Loan_Applications,
    SUM(loan_amount)        AS Total_Funded_Amount,
    SUM(total_payment)      AS Total_Amount_Received
FROM bank_loan_data
GROUP BY home_ownership
ORDER BY home_ownership;
```

Home_Ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
MORTGAGE	17198	219329150	238474438
NONE	3	16800	19053
OTHER	98	1044975	1025257
OWN	2838	29597675	31729129
RENT	18439	185768475	201823056

5. Exploratory Insights

Based on SQL analysis, the following insights can be derived:

- Loan applications show monthly variation, indicating seasonality in borrowing behavior
- Higher loan amounts are often associated with higher risk categories
- Borrowers with higher DTI ratios are more likely to default
- Certain purposes (e.g., debt consolidation) dominate loan applications

- Employment length and home ownership influence repayment behavior

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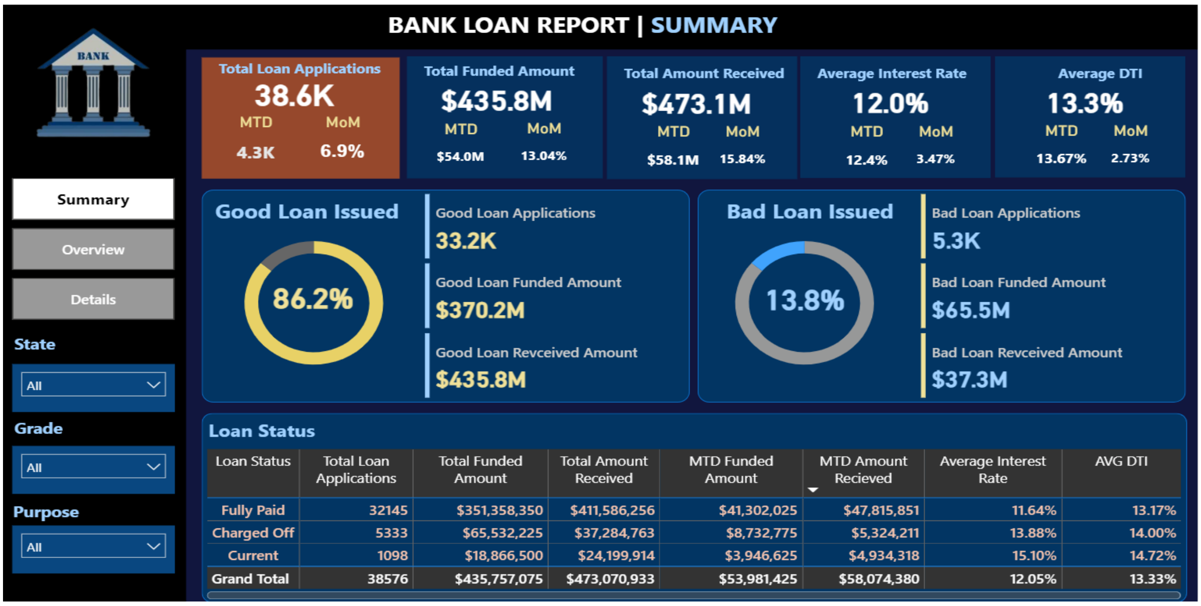
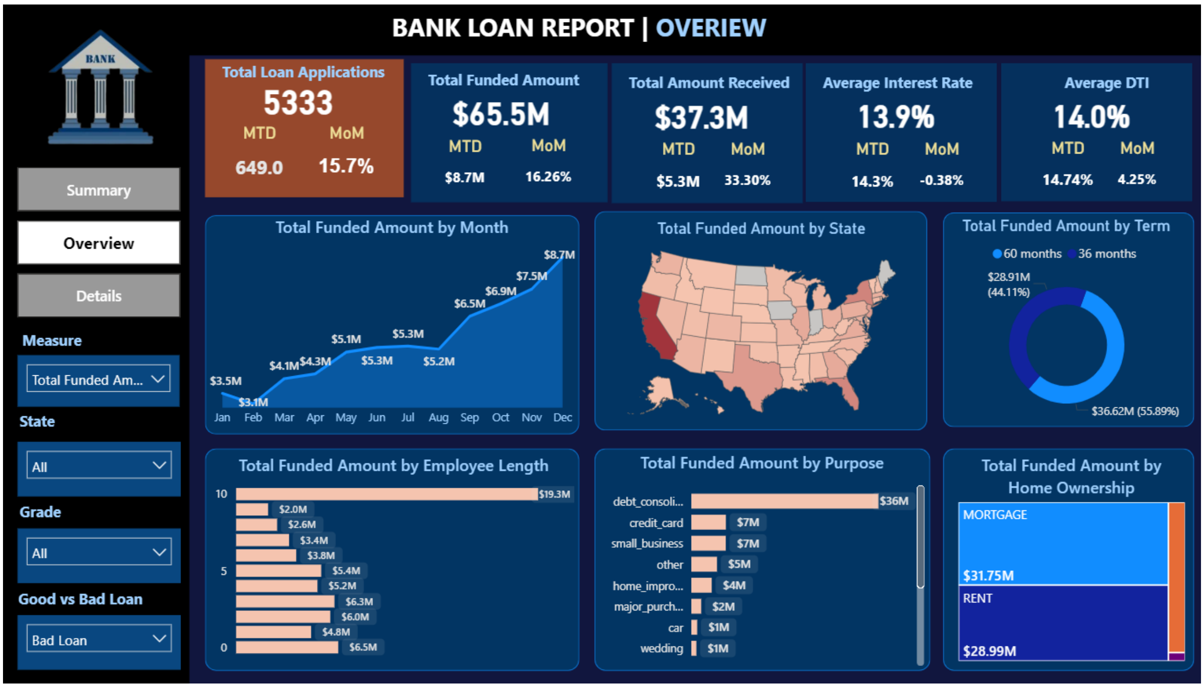
6. Dashboard (Power BI)





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7. Business Recommendations

Based on the analysis, the following recommendations are proposed:

Risk Management

- Implement stricter approval criteria for high DTI borrowers
- Monitor “Charged Off” loans to identify early warning signals

Lending Strategy

- Focus on low-risk borrower segments (stable employment, lower DTI)
- Optimize interest rates based on borrower risk profiles

Portfolio Optimization

- Diversify loan portfolio across purposes and regions
- Reduce exposure to high-risk loan categories

Operational Improvements

- Improve data collection for better risk modeling
- Automate loan monitoring using dashboards